CCEE 543	Mobile Communications	3(3,0)
This course deals with the evolution of cellular technologies from 2G to LTE: cellular systems; medium access techniques; wireless standardization; GSM networks (history, architecture overview, access network, GSM air interface, dimensioning aspects, network design and planning, handover, call handling, network operation); 3G mobile systems architecture, 3G air interface, protocols and procedures used in 3G, 3G functionalities such as: idle mode camping, power control and capacity management, handover in 3G; introduction to 4G/LTE. **Co-requisite**: CCEE 449.		

CCEE 562	Artificial Intelligence	3(3,0)
This course introduces students to the fundamentals of knowledge representation and learning in artificial intelligence. The emphasis will be on understanding fundamental concepts in artificial intelligence and applying corresponding approaches to solve practical problems and develop useful software applications. Covered topics include intelligent agents, informed and uninformed search strategies, and adversarial search. **Prerequisite**: CCEE 214 or equivalent.		

CCEE 564	Machine Learning and Data Mining	3(3,0)
This course introduces students to the basic knowledge representation and learning techniques. The emphasis is on understanding the data mining process and applying the corresponding approaches to solve practical problems and develop intelligent software applications. The course covers several topics that lie within classification, prediction, and clustering. **Prerequisite**: CCEE 214. Equivalent to COSC 480.		

CCEE 595A	Summative Learning Project 1	1(1,0)
The first phase of a team-oriented, project-based experience culminates in the creation of an artifact; milestones include project selection and proposal, a creative solution, a report, a presentation, and a demonstration of the device. **Prerequisites**: ENGL 217, Senior Standing.		

CCEE 595B	BE Summative Learning Project 2	3(3,0)
The second phase of a team-oriented, project-based experience that culminates in the creation of an artifact; milestones include: project selection and proposal, a creative solution, a report, a presentation, and a demonstration of the created device. **Prerequisite:** CCEE 595A.		

II. Elective Courses

Students may meet their engineering technical electives requirements from the following selection of elective courses.

CCEE 457	Cybersecurity Fundamentals	3(3,0)
This course equips learners with workforce preparedness skills and establishes a foundation for success in cybersecurity. Participants acquire, apply, and refine cybersecurity knowledge and skills through a sequence of comprehensive, practical experiences that solidify their learning. This course imparts fundamental cybersecurity		

concepts and abilities at the introductory level, encompassing threat reduction, defense strategies, and post-incident forensics. **Prerequisite: CCEE 354**

CCEE 498	Special Topics in Computer & Communication Engineering	3(3,0)
This course is designed to enable students to study current special topics of interest, carefully selected from CCE-related topics. The contents of such a course are to be determined by the instructor and the department.		

CCEE 511	Mobile Application Development	3(3,0)
Today's applications are increasingly mobile. Computers are no longer confined to desks and laps; they live in our pockets and hands. This course teaches students how to build mobile apps for Android and iOS, two of today's most popular platforms, and how to deploy them in the Android Market and the App Store. Students learn how to write native apps for Android using Eclipse and the Android SDK, for iPhones, iPod touches, and iPads using Xcode and the iOS SDK, and for web apps on both platforms. **Prerequisite:** CCEE 214. Equivalent to COSC 435.		

CCEE 513	Operating Systems	3(3,0)
This course covers operating systems principles; scheduling and resource management; virtual memory; file systems; concurrent processing and synchronization; security and protection; the Internet; network structures; distributed operating systems; Web technologies; and operating systems (URL, HTML, HTTP, applets). A case study of a UNIX operating system is covered during the course. **Prerequisite**: CCEE 214. Equivalent to COSC 451.		

CCEE 514	Advanced Web Programming	3(3,0)
This course focuses on server-side programming. It teaches students how to connect their website or web application to a database and save and retrieve data from it. The course exposes students to web controls, validation controls, data source controls, data binding controls, state management, and working with third-party media such as XML and web services. **Prerequisite**: CCEE 411/COSC 333. Equivalent to COSC 434.		

CCEE 515	Distributed Computing	3(3,0)
This course will introduce students to the challenges faced in constructing parallel and distributed applications, including testing, debugging, and performance evaluation. The student will be trained in various implementation techniques, paradigms, architectures, and programming languages. **Prerequisite**: CCEE 528 or equivalent.		

CCEE 516	Advanced Programming and Database Management Systems	3(2,2)
This course will introduce students to managing their own databases and querying them, as well as to managing Data Warehousing. Students will also learn advanced programming tools, including Bigtable, NoSQL, R, Python, SCALA, MapReduce, Elasticsearch, and apply these tools to address big data issues. **Prerequisite**: CCEE 315 or equivalent.		

CCEE 520	Parallel Processing	3(3,0)
This course introduces students to various aspects of parallel processing, including parallel architectures, algorithms, systems, programming languages, and implementations. Students will be expected to work with recent existing parallel machines and design and implement parallel processing projects. **Prerequisite**: CCEE 221.		

CCEE 521	Hardware-Software Co-Design	3(3,0)
This course introduces the student to the design of complex embedded systems. Several models and methods are presented to help the student design software and hardware components that work together. Models to describe hardware and software components are introduced (specifications). Then, hardware-software partitioning and design space exploration are introduced as part of the system design. Performance analysis and estimation techniques are then explained. Finally, a current hardware-software co-design topic is chosen and introduced. Students will be expected to design and implement a project using the information in this course. **Prerequisite**: CCEE 221.		

CCEE 535	Digital Signal Processing	3(3,0)
This course focuses on digital signal processing (DSP) and its applications. It helps students understand how digital signal processing can be applied in operational systems. Students are required to develop simulations of the learned concepts using MATLAB. Specific topics covered include a review of mathematical tools used in DSP, digital filter structure, digital filter design, simple DSP algorithm implementation, and spectral analysis of signals. **Prerequisite**: CCEE 331.		

CCEE 536	Digital Image Processing	3(3,0)
Two-dimensional signals and systems. Image formation and perception. Representation, coding, filtering, restoration, and enhancements. Feature extraction and scene analysis. Introduction to computer vision. Introduction to Medical Imaging. **Prerequisite:** CCEE 331 or Equivalent.		

CCEE 540	Satellite Communication Systems	3(3,0)
This course focuses on the fundamentals of satellite technology. Topics include the history of satellites; satellite missions and applications; the study of satellite orbits and trajectories; satellite launch and in-orbit operations; satellite link fundamentals; satellite hardware; Communication Satellites; GPS; and Direct Broadcast Satellites. **Prerequisite**: CCEE 447.		

CCEE 542	Microwave Communications	3(3,0)
This course introduces the fundamental principles of microwave circuit analysis and design. Topics include review of electromagnetic theory, Transmission Line Theory, Impedance Matching and Tuning, Microwave Network Analysis, Power Dividers, and Microwave Propagation. This course also includes some laboratory experiments on microwave circuits—prerequisite: ELEC 340.		

CCEE 545	Advanced Mobile Communications	3(3,0)
This course seeks to provide insight into the development of the latest generations of mobile networks. In this course, we will focus mainly on LTE and LTE-Advanced (LTE-A): Motivation, architecture, air interface, radio procedures, quality of service, and planning. In addition, we will briefly discuss 5G networks and the Internet of Things. **Co-requisite**: CCEE 449.		

CCEE 546	Array Processing	3(3,0)
This course includes an introduction to Mono- and Multi-Antenna Concepts: multi-antenna benefits; beamforming techniques; adaptive antennas; MIMO systems: diversity/multiplexing tradeoff; multi-antenna transmission techniques (Alamouti and spatio-temporal techniques); multi-antenna reception techniques; transmission strategies; applications (MIMO in 3G and LTE). **Prerequisite**: CCEE 444.		

CCEE 547	Optical Communications	3(3,0)
This course is designed to progress from the description of the components in a fiber link to the interconnections within a link or a network. The components of fiber links will be discussed (fiber, connections, sources, and receivers). These optical components will be integrated into a complete optical link. **Prerequisite**: CCEE 331.		

CCEE 552	Network Security	3(3,0)
Network security is the protection of the underlying networking infrastructure from unauthorized access, misuse, or theft. It involves creating a secure infrastructure for devices, applications, and users to operate securely. **Pre-requisite**: CCEE 354 or COSC 360.		

CCEE 553	Advanced Communication Networks	3(3,0)
In this course, we will explore current network architectures and topologies, such as telephone systems (PDH/SDH, SONET), Frame Relay, ATM, and MPLS, as well as technologies under development. We will also learn how to assess network performance and which technologies are used to enhance service quality. At the end of the course, the student will be able to identify the technologies to be used in a given environment and study its performance. **Prerequisite**: CCEE 354.		

CCEE 554L	Computer Networks Lab 2	1(0,2)
This Lab helps prepare students seeking to pass advanced Cisco Certifications (CCNA 1). The student will acquire knowledge of network equipment and protocol functionality, learn to build a simple Ethernet network using routers, switches, and computers, and use router CLI commands to perform basic configuration and verification. The student will also learn valuable network problem-solving techniques and concepts through hands-on skills using hardware devices. **Co-requisite**: CCEE 354L.		

CCEE 555	Advanced Computer Networks	3(3,0)
This is an advanced networking course that requires active student participation. In this course, we will explore TCP; Internet inter-domain routing; Internet quality of service and traffic engineering; Internet traffic measurement and analysis; data		

centers; the analysis and performance of content distribution networks; and Software-defined networking. In addition, the student will learn how to build client/server computer network applications using sockets. **Prerequisite**: CCEE 354.

CCEE 555L	Computer Networks Lab 3	1(0,2)
This Lab helps prepare students seeking to pass the Cisco Routing and Switching Essentials (CCNA 2) Certification. The primary focus of this Lab is routers and switches in small networks. The student will learn the architecture, components, and operations of routers and switches, as well as their basic configuration and functionality. They will also learn how to configure and troubleshoot static and dynamic routing protocols, access-control lists, VLANs, DHCP for IPv4 and IPv6 networks, and NAT operations. **Prerequisite**: CCEE 354L.		

CCEE 556	Network and System Administration	3(3,0)
This course will help students become system administrators and install, configure, maintain, and troubleshoot a Microsoft Windows Server Operating system. It provides them with the knowledge and skills to work with Active Directory technologies in environments including complex network services. This course also provides students with basic knowledge of installing, configuring, and maintaining Linux operating systems. By the end of this course, students will have all the required materials to submit and pass the MCSA: Windows Server certification exam. **Prerequisite**: CCEE 354.		

CCEE 556L	Computer Networks Lab 4	1(0,2)
This Lab helps prepare students seeking to pass the Cisco Routing and Switching Essentials (CCNA 3) Certification. The primary focus of this Lab is routers and switches in large and complex networks. The student will learn how to configure routers and switches with advanced functionalities. They will also learn how to configure and troubleshoot enhanced switching technologies, first-hop redundancy protocols in a switched network, wireless routers and clients, and routers in complex networks. Finally, they will learn how to manage CISCO IOS software licensing and configuration files. **Prerequisite**: CCEE 555L.		

CCEE 561	Computer Vision	3(3,0)
This course introduces the principles, models, and applications of computer vision. The course will cover image structure, projection, stereo vision, and the interpretation of visual motion. Case studies of industrial (robotic) applications of computer vision, including visual navigation for autonomous robots, robot hand-eye coordination, and novel man-machine interfaces. **Prerequisite**: CCEE 214 or COSC 214.		
CCEE 563	**Robotics**	**3(3,0)**
Introductory historical development of robotics; rigid-object kinematics; robot arm kinematics; inverse kinematics; dynamics; introduction to trajectory planning and manipulator control. **Prerequisite**: MATH 311 and MATH 314. Equivalent to MECA 524.		

CCEE 566	Natural Language Processing	3(3,0)
This course introduces students to the field of natural language processing (NLP). The student is first introduced to word and sentence tokenization. The student then uses the learned skills to implement systems for text classification and sentiment analysis, spelling correction, information extraction, parsing, meaning extraction, and question answering, Machine learning algorithms as well as algorithms like n-gram language modeling, naive bayes and maxent classifiers, sequence models like Hidden Markov Models, probabilistic dependency and constituent parsing, and vector-space models of meaning will be introduced as needed for the above NLP applications. **Prerequisite**: CCEE 214, COSC 214, or Equivalent.		

CCEE 567	Optimization	3(3,0)
This course introduces students to the theory, algorithms, and applications of optimization. The optimization methodologies include linear programming, network optimization, integer programming, and decision trees. It includes a team project in which students select and solve a problem in practice. **Prerequisite**: MATH 215 or equivalent. Equivalent to: BADM 420 and MECH 571.		

CCEE 568	Big Data and Analytics/Big Data System	3(3,0)
This course enables students to understand why the Big Data Era has emerged. Students will become conversant in the terminology and core concepts of big data problems, applications, and systems. Students will learn how to make Big Data useful in their business or career. Students will be introduced to one of the most common frameworks, Hadoop, that has made big data analysis easier and more accessible -- increasing the potential for data to transform our world. **Prerequisite**: CCEE 564 or equivalent.		

CCEE 597	Advanced Topics in Computer & Communications Engineering	3(3,0)
This course is designed to enable students to study a selected advanced topic of interest, carefully chosen from the CCE engineering-related topics. The content outline of such a topic is to be determined by the instructor and approved by the department. **Prerequisite**: Instructor's consent.		